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PART II: EFFECTS OF PROBE COMPENSATION — UNDERSTANDING PROBES

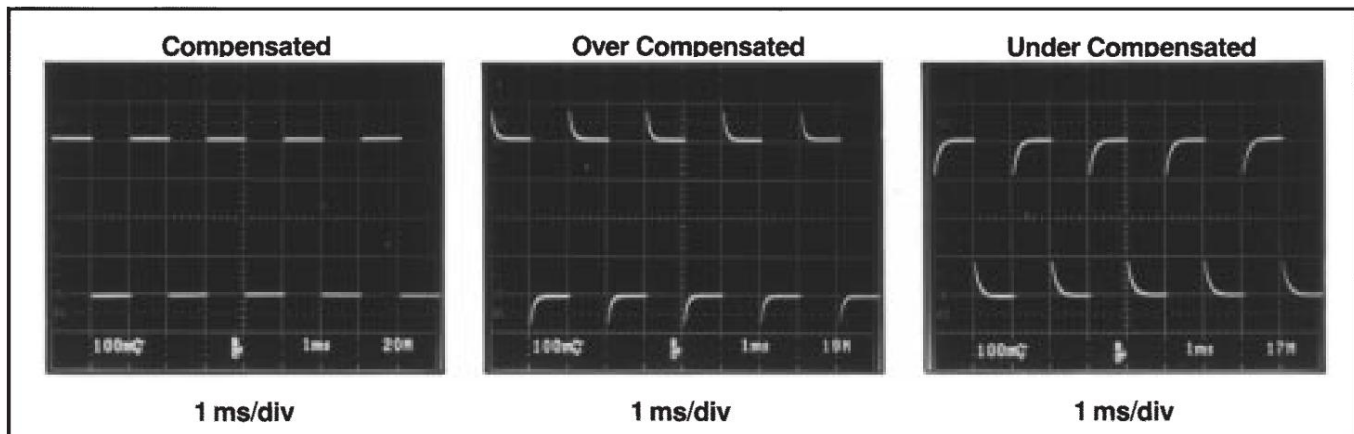


Figure 2-1. Shows the display associated with correctly and incorrectly compensated probes.

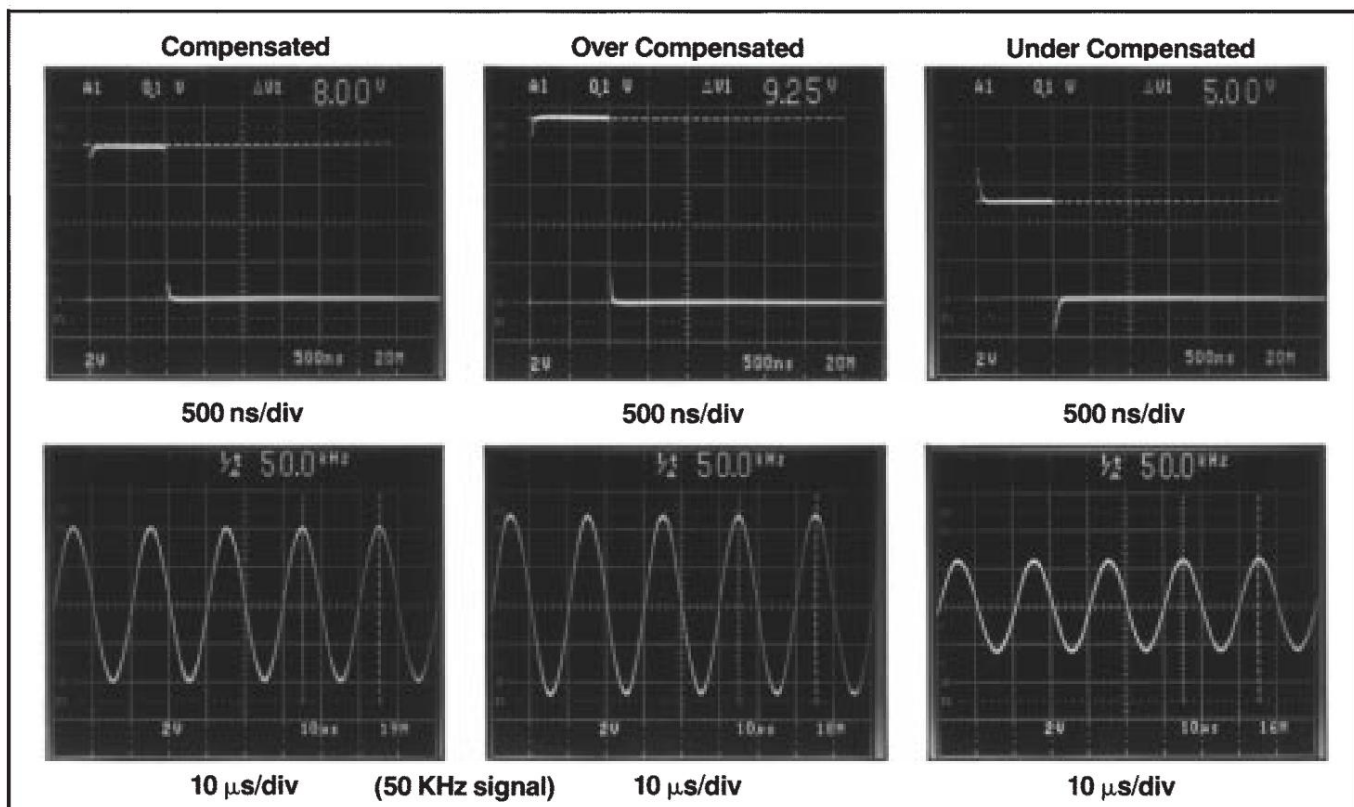


Figure 2-2. Shows the effects on faster pulses and sinewaves when an incorrectly compensated probe is used. Note that the much faster sweep rates used to correctly view these waveforms does not warn the user of an adjustment problem.

Tips on using probes

Compensating the probe. The most common mistake in making scope measurements is forgetting to compensate the probe. Improperly compensated probes can distort the waveforms displayed on the scope. The probe should be compensated as it will be used when you make the measurement.

The basic low frequency compensation (L.F. comp.) procedure is simple:

- Connect the probe tip to the scope CALIBRATOR (refer to Scope Calibrator Outputs.)
- Switch the channel 1 input coupling to dc.
- Turn on the scope and move the CH1 VOLTS/DIV switch to pro-

duce about four divisions of vertical display.

- Set the sweep rate to 1mSec/div. (for line-driven calibrators see Scope Calibrators Outputs.)
- Use a non-metallic alignment tool to turn the compensation adjust until the tops and bottoms of the square-wave are flat.

PART III: ADVANCED PROBING TECHNIQUES

Introduction:

In Part III we will examine some of the more advanced probing techniques associated with accessing high frequency and complex signals, such as fast ECL, waveforms offset from ground, and true differential signals.

Most of the techniques to be described follow recommended practices outlined throughout this Booklet, and to a large extent involve proper grounding techniques.

Workers in the audio and relatively low frequency fields may wonder what all the fuss is about, and may comment "I don't have any of these problems," or "I can't see any difference when I use different ground lead lengths, or even when I leave the ground lead completely off?"

In order to see aberrations caused by poor grounding techniques, two conditions must exist:

1. The scope system bandwidth must be great enough to handle the high frequency content existing at the probe tip.
2. The input signal must contain enough high frequency information (fast risetime) in order to cause ringing and aberrations due to poor grounding techniques.

To illustrate these points, a 20 MHz scope was used to access a 1.7 nS pulse by using a standard passive probe with a 6" ground lead.

NOTE: A fast scope can be made into a slow scope simply by pushing the Bandwidth Limit (B/W Limit) button?

We used a 350 MHz scope with a 20 MHz B/W Limit function.

Figure 3-1 shows the resultant clean displayed pulse with a risetime of about 20 nS (17.5 MHz).

This display does not represent conditions actually existing at the probe tip, because the 20 MHz measurement system cannot "see" what's really happening.

Figure 3-2 shows what the probe tip signal really looks like when a 350 MHz scope is used under the same conditions (B/W Limit off).

The observed risetime has improved to about 2 nS, but we have serious problems with ringing and aberrations, caused by incorrect grounding techniques.

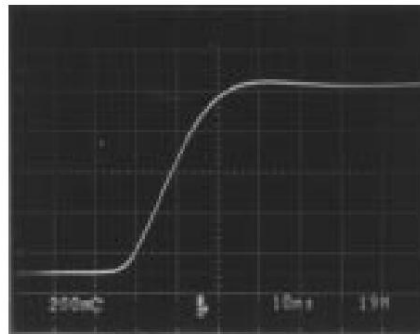


Figure 3-1. Resultant clean, but incorrect display caused by inadequate scope system bandwidth.

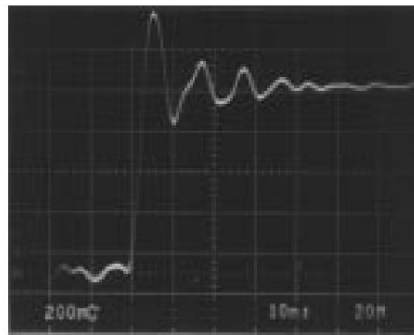


Figure 3-2. The same input signal as shown in figure 3-1, but accessed by a 350 MHz system bandwidth scope (same 6" ground lead).

The problem can now be seen because the scope system bandwidth is great enough to pass and display all the frequency content existing at the probe tip.

To further stress the points about high frequency content and scope system bandwidth, let's assume an input pulse with a risetime of about 20 nS. If the signal is accessed by the same probe /6" ground lead /350 MHz system, it would look very much like the display in figure 3-1.

There would be no frequency content higher than 17.5 MHz (20 nS Tr). The 6" ground lead would not ring, and would therefore be the correct choice for accessing this relatively slow signal.

In the following sections we discuss how to recognize signal acquisition problems, and how to avoid them.

Techniques for probing ECL, high speed 50 Ω environments, and accessing true differential signals are also discussed.

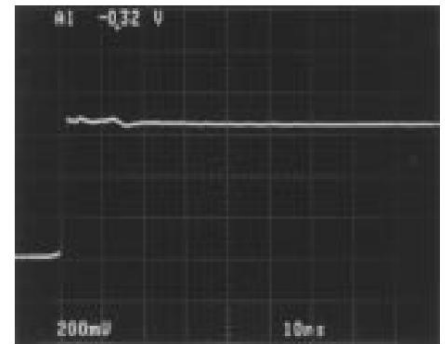


Figure 3-3. 1 nS Tr pulse accessed via an ECB to Probe Tip Adaptor (test point)

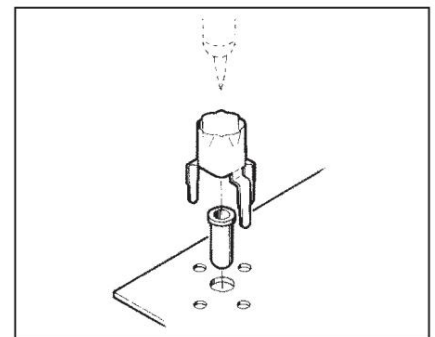


Figure 3-4. Typical ECB to Probe Tip Adaptor installation

Probe Ground Lead Effects. In Part I we discussed the basic need for probe grounding, and showed several different ways of looking at the effects of correct, and incorrect probe grounding.

In this section, we will expand upon these techniques and show how to identify problem areas.

When a probe (high Z, low Z, passive or active) is connected to the circuit under test via an ECB to Probe Tip Adaptor (test point), the coaxial environment existing at the probe tip is extended through the adaptor to the signal pick-off point, and to the ECB ground plane (or device ground).

Figure 3-3 shows what a typical 1 nS Tr pulse looks like when a suitable probe is connected to the circuit via an ECB to Probe Tip Adaptor.

Figure 3-4 shows a typical ECB to Probe Tip Adaptor (test point) installation.

These test points are available in three sizes to accept miniature, compact or sub-miniature series probes.

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If a flexible ground lead is used in place of the ECB to Probe Tip Adaptor, the 1 nS Tr input step (with high frequency content up to 350 MHz) will cause the ground lead to ring at a frequency determined by the ground lead inductance and the probe tip and source capacitance.

Figure 3-5 shows the effect of using a 6" ground lead to make the ground connection.

The ring frequency for the 6" ground lead/probe tip C combination is 87.5 MHz. This signal is injected in series with the wanted signal and appears at the probe tip, as shown in figure 3-6.

Unfortunately, the problem is not this simple.

The probe's coaxial environment has been disrupted at the signal acquisition point by ground lead inductance, and is no longer correctly terminated (for high speed signal acquisition).

This abrupt transition leaves the probe's outer shield susceptible to ring frequency injection (the ground lead inductance is in series with the outer braid)

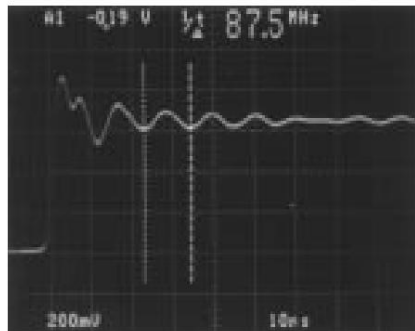


Figure 3-5. Effect of a 6" ground lead on a 1 nS Tr input step.

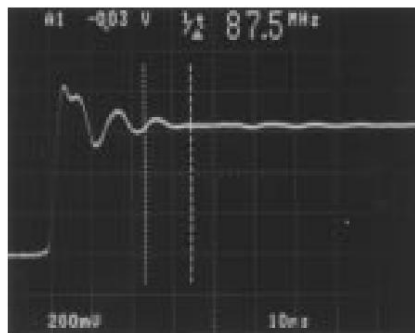


Figure 3-7. The same setup as in figure 3-5, except that the probe cable has been repositioned, and a hand has been placed over part of the probe cable.

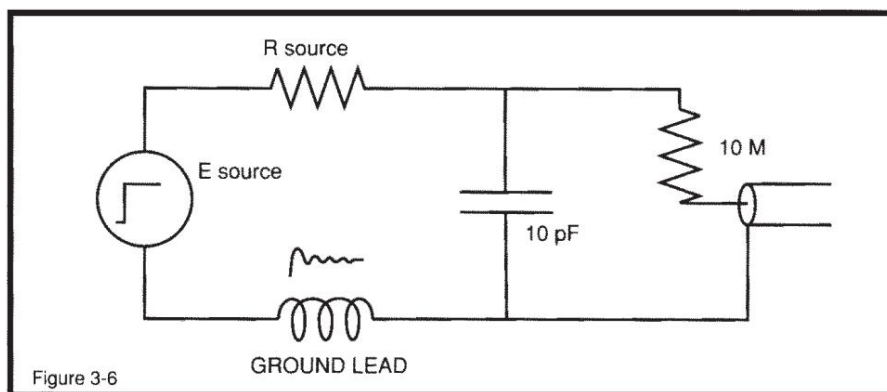


Figure 3-6. Equivalent circuit, ground lead inductance (excess inductance).

The now unterminated probe cable system develops reflections, which intermix with the ring frequency and the signal to produce a multitude of problems and unpredictable results.

Herein lies the key to the identification of ground lead problems.

Figure 3-7 shows exactly the same setup as in figure 3-5, except that the probe cable has been moved, and a hand has been placed over part of the probe cable.

KEY: If touching or moving the probe cable produces changes in the display, you have a probe grounding problem.

A correctly grounded (terminated) probe should be completely insensitive to cable positioning or touch.

Ground Lead Length. All things being equal, the shortest ground lead produces the highest ring frequency.

If the lead is very short, the ring frequency might be high enough to be outside the passband of the scope, and/or the input frequency content may not be high enough to stimulate the ground lead's resonant circuit.

In all cases, the shortest ground lead should be used, consistent with the need for probe mobility.

If possible, use 3" or shorter ground leads, such as the Low Impedance Contact (Z Lead). These are supplied with the Tektronix P613X and P623X family of probes.

One final note. The correct probe grounding method depends on the signal's high frequency content, the scope system bandwidth, and the need for mobility between test points.

A 12" ground lead may be perfect for many lower frequency applica-

tions. It will provide you with extra mobility, and nothing will be gained by using shorter leads.

If in doubt, apply the cable touch test outlined previously.

Ground Loop Noise Injection.

Another form of signal distortion can be caused by signal injection into the grounding system.

This can be caused by unwanted current flow in the ground loop existing between the common scope and test circuit power line grounds, and the probe ground lead and cable.

Normally, all these points are, or should be at zero volts, and no ground current will flow.

However, if the scope and test circuit are on different building system grounds, there could be small voltage differences, or noise on one of the building ground systems.

The resulting current flow (at line frequency or noise frequency) will develop a voltage drop across the probe cable's outer shield, and be injected into the scope in series with the desired signal.

Inductive Pickup in Ground Loops. Noise can enter a common ground system by induction into long 50 Ω signal acquisition cables, or into standard probe cables.

Proximity to power lines or other current-carrying conductors can induce current flow in the probe's outer cable, or in standard 50 Ω coax. The circuit is completed through the building system common ground.

Prevention of Ground Loop Noise Problems. Keep all signal acquisition probes and/or cables away from sources of potential interference.

Verify the integrity of the building system ground.

If the problem persists, open the ground loop:

1. By using a Ground Isolation Monitor like the Tektronix A6901.
2. By using a power line isolation transformer on either the test circuit or on the scope.
3. By using an Isolation Amplifier like the Tektronix A6902B.
4. By using differential probes (see Differential Measurements).

NOTE: Never defeat the safety 3-wire ground system on either the scope or on the test circuit.

Do not "float" the scope, except by using an approved isolation transformer, or preferably, by using the Tektronix A6901 Ground Isolation Monitor.

The A6901 automatically reconnects the ground if scope ground voltages exceed ± 40 V.

Induced Noise in Probe Ground Leads. The typical probe ground lead resembles a single-turn loop antenna when it is connected to the test circuit.

The relatively low impedance of the test circuit can couple any induced voltages into the probe, as shown in figure 3-8.

High speed logic circuits can produce significant electro-magnetic (radiated) noise at close quarters.

If the probe ground lead is positioned too close to certain areas on the board, interference signals could be picked up by the loop antenna formed by the probe ground lead, and mix with the probe tip signal.

Question: Is this what my signal really looks like?

Moving the probe ground lead around will help identify the problem.

If the **noise level** changes, you have a ground lead induced noise problem.

A more positive way of identification is to disconnect the probe from the signal source and clip the ground lead to the probe tip.

Now use the probe/ground lead as a loop antenna and search the board for radiated noise.

Figure 3-9 shows what can be found on a logic board, **with the probe tip shorted to the ground lead.**

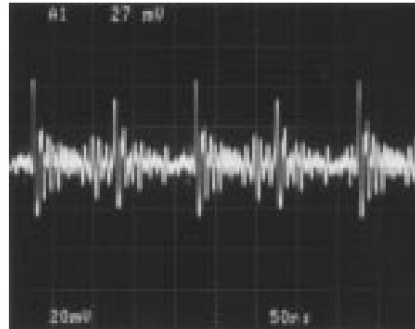


Figure 3-9. Induced noise in the probe ground loop (tip shorted to the ground clip).

This is radiated noise, induced in the single-turn loop and fed to the probe tip.

The significance of any induced or injected noise increases with reduced working signal levels, because the signal to noise ratio will be degraded. This is especially true with ECL, where signal levels are 1 V or less.

Prevention: If possible, use an ECB to Probe Tip Adaptor (test point). If not, use a Z Lead or short flexible ground lead.

Also, bunch the ground lead together to make the loop area as small as possible.

Bias Offset Probes. A Bias/Offset probe is a special kind of Low Z design with the capability of providing a variable bias or offset voltage at the probe tip.

Bias/Offset probes like the Tektronix P6230 and P6231 are useful for probing high speed ECL circuitry, where resistive loading could upset the operating point.

They are also useful for probing higher amplitude signals (up to ± 5 V), where resistive loading could affect the DC level at some point on the waveform.

Bias/Offset probes are designed with a tip resistance of 450Ω (10X). When these probes are connected into a 50Ω environment, this loading results in a 10% reduction in peak to peak source amplitude. This round-figure loading is more convenient to handle than that produced by a standard 500Ω (10 X) Low Z probe, which would work out at 9.09% under the same conditions.

It is important to note that bias/offset probes always present a 450Ω resistive load to the source, regardless of the bias/offset voltage selected.

The difference between bias/offset and standard Low Z probes lies in their ability to null current flow **at some specific and selectable point** on the input waveform (within ± 5 V).

To see how bias/offset probes work, let's take a typical $10 \times 500 \Omega$ Low Z probe and connect it in the circuit shown in figure 3-10.

By taking a current flow approach we find that at one point on the waveform the source voltage is -4 V, therefore the load current will be;

$$I = ER / (R_s + R_p + R_{\text{scope}}) = 4 / 550 = 7.27 \text{ mA.}$$

Therefore the voltage drop across the 50Ω source resistance (R_s) will be;

$$E = IR = 7.27 \times 10^{-3} \times 50 = 0.363 \text{ V}$$

And the measured pulse amplitude will be $-4 - 0.363 = -3.637$ V (E dut), or about 9% down from its unloaded state.

If we substitute the 500Ω Low Z probe with a 450Ω bias/offset probe, the circuit will look like figure 3-11.

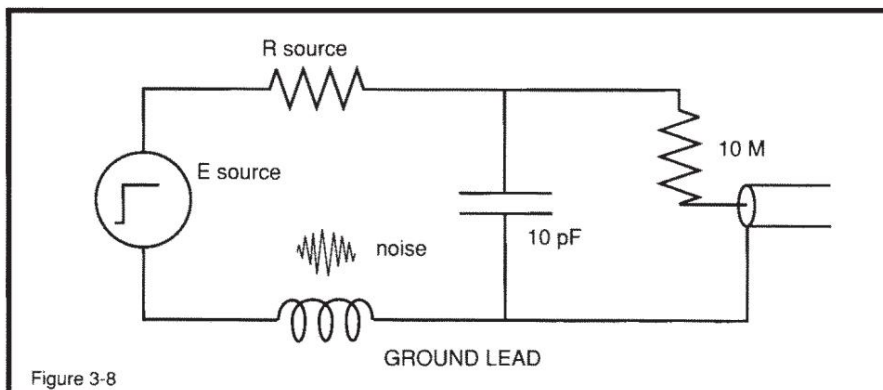


Figure 3-8

Figure 3-8. Equivalent Circuit. Ground Lead Induced Noise.

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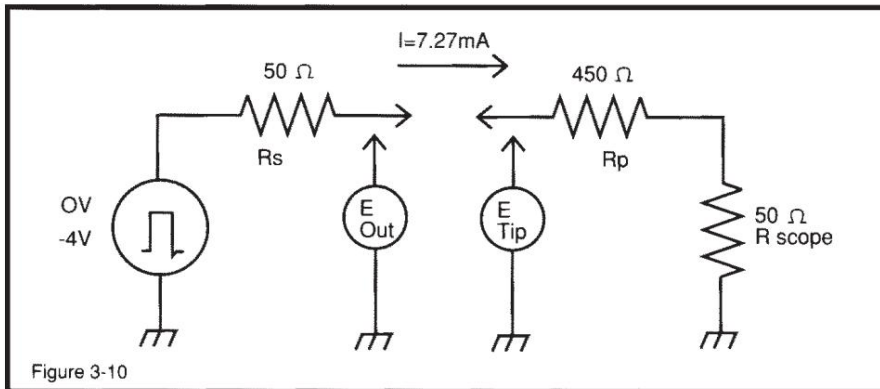


Figure 3-10. Low Z 10X 500 Ω probe connected to a 50 Ω source.

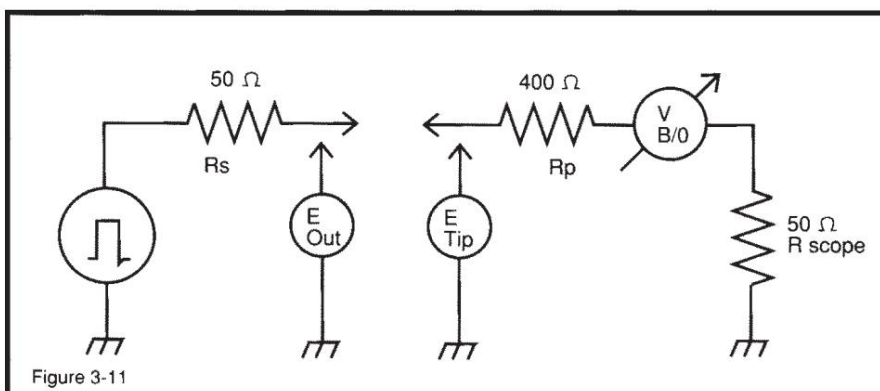


Figure 3-11. A 450 Ω Bias/Offset probe connected to a 50 Ω source.

With the bias/offset adjusted for 0 V, the effect on the circuit will be similar to a 500 Ω Low Z probe, except for the small resistive change.

Figure 3-12 shows the source waveform acquired by a 10 M Ω probe.

Figure 3-13 shows the effect on the waveform when the 450 Ω probe is added.

As expected, the pulse amplitude has been reduced from -4 V to 3.60 V, or exactly 10% down.

Figure 3-14 shows the effect of adjusting the offset to -4 V. The -4 V bias opposes the signal at the -4 V level and results in zero current flow, and the source is effectively unloaded **at this point**.

However, when the signal returns to ground level, there is a 4 V differential between the top of the pulse and the bias/offset source. Current will flow, and Ohms Law will dictate that the top of the pulse will go negative by -40 mV (10%).

Sometimes it is desirable to adjust the offset mid-way between the peak to peak excursions. This distributes the effect of resistive loading between the two voltage swings.

Figure 3-15 shows the effect of adjusting the bias/offset to -2 V. Current flow will be the same for both signal swings, and they will be equally down by 5%, for a total of 10%.

Summary:

1. Bias/Offset probes can be adjusted (within ± 5 V) to provide zero resistive (effective) loading at one selected point on the input waveform.
2. Bias/Offset probes can be used to simulate the effect of pull-up or pull-down voltages (within ± 5 V) on the circuit under test (voltage source impedance is 450 Ω).
3. Bias/Offset probes always present a total resistive load of 450 Ω , and reduce the peak to peak amplitude of 50 Ω sources by 10%.
4. For simplicity, we have ignored the effects of capacitive loading. Typically, Bias/Offset probes have less than 2 pF tip C.

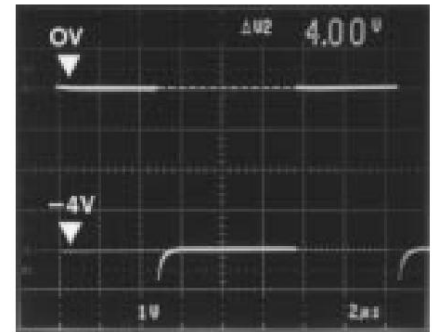


Figure 3-12. Unloaded negative-going 4 V pulse acquired by a 10 M Ω probe.

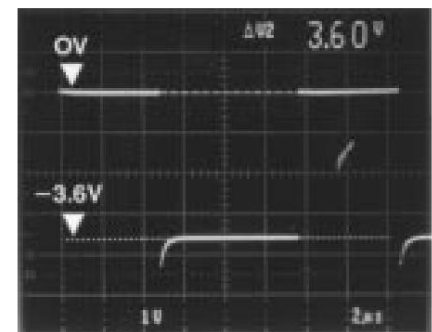


Figure 3-13. Effect of connecting a 450 Ω Bias/Offset probe (offset = 0 V). Minus level has been reduced by 10%.

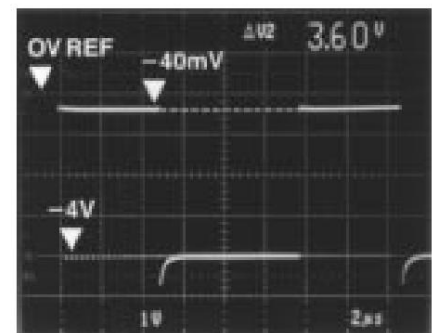


Figure 3-14. Bias/Offset adjusted for -4 V. Signal current at the -4 V level is zero. Current flow at ground level is maximum. Peak to peak amplitude remains the same (10% down).

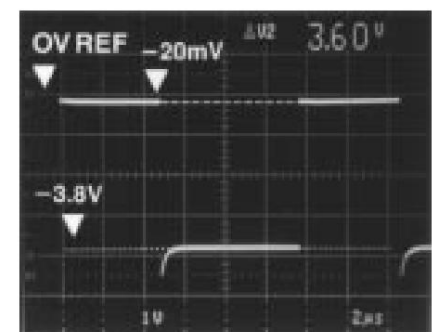


Figure 3-15. Bias/Offset adjusted for -2 V. Load current distributed between the negative and positive-going swings. Peak to peak amplitude remains the same (10% down).